

Response to Reviewers

Manuscript CHB-D-26-04350

At the Point of Entry: Receiving Help and Newcomer Contribution on Stack Overflow

We thank the editor and both reviewers for their careful and constructive reading of the manuscript. In preparing this revision we substantially re-analysed the data: we re-matched treated and control questions using nearest-neighbor propensity-score matching *with replacement* (caliper 0.05), with exact strata on calendar year, primary tag, and the number of tags, and re-fit every Cox model on the resulting sample, so that all reported estimates derive from a single, consistent pipeline run. The principal changes are: (i) matched-pair bootstrap confidence intervals for the treatment effect — pooled and, because it carries the paper’s headline, for the newcomer stratum at 500 replicates — replacing the earlier claim that clustering was unnecessary; (ii) a full decomposition of the helping outcome by help type (answers, comments, edits, and their composites), which retains answers — now explicitly justified on construct grounds — as the headline outcome and shows the reciprocity signal is answer-borne; (iii) a flexible, non-parametric response-time specification (discrete bins), re-estimated with observable answer-quality controls; (iv) new robustness analyses for observable selection, entry-cohort heterogeneity, and left-truncation/churn; (v) a more modest, tenure-focused framing throughout, under which the pooled association is small, the newcomer gradient is the most robust pattern, the response-time result is offered as support for an immediacy account, and the quality-controlled estimates are read as a conservative post-treatment bound; (vi) a shortened Theory section that is linked directly to the empirical design, with interpretation-relevant appendix material promoted into the main text; (vii) a restructured Discussion, in which the Limitations section is reorganized around what we measure, what we can identify, what we cannot see, and where the estimates apply, the generative-AI material is gathered into its own subsection rather than scattered across the section, and the common-support accounting is moved to Appendix Appendix C; and (viii) a full language and consistency edit of the revised manuscript. One further transparency item: a legacy supplementary analysis of lifetime contributions was removed from the appendix (see our response to R1 Q8), because it derived from a superseded sample and constructs not defined in the current pipeline; no claim in the revised manuscript relies on it. Analysis code and step-by-step instructions to reproduce every table and figure are provided in the replication package described in the Data availability statement.

Editor requirements

- **Point-by-point response:** attached (this document), covering every Reviewer 1 sub-point (Q1–Q9), every Reviewer 2 point (1–9), and each editor requirement.
- **Editable source files:** the revised manuscript is uploaded in editable source form alongside the PDF, per the production requirement.
- **Highlights:** a separate **Highlights** file accompanies the submission with four bullet points, each verified at or below 85 characters including spaces, worded consistently with the title and the revised abstract.
- **Graphical Abstract:** we respectfully decline the optional Graphical Abstract.

For each point below we restate the reviewer’s comment (in the left column) and describe our response and the corresponding changes to the manuscript (right column), with explicit references to the revised sections and tables.

Response to Reviewer 1

Reviewer comment	Response and changes to the manuscript
(Q1) “The authors should separate the empirical question from the proposed psychological mechanism. The data can show whether users answer more after receiving help, but they do not directly measure gratitude, moral obligation, or incentive displacement.” (Q5) “The ‘re-engagement window’ interpretation is also plausible, but the paper does not observe sessions, exits, returns, or notification behavior. This should be presented as a possible explanation rather than as an established mechanism.” (Q7) “The authors should acknowledge that they do not directly observe psychological mechanisms such as gratitude, obligation, or status motivation.”	We reframed the manuscript so the estimand is stated plainly while every psychological and session mechanism is marked as a <i>proposed</i>, unmeasured interpretation; no models were re-fit for this change. The abstract states that “our data identify short-run behavioral patterns, not the psychological mechanisms (gratitude, moral obligation, incentive displacement) that theory proposes,” and the Introduction’s opening paragraph states that “data do not directly measure the psychological mechanisms that theory proposes (gratitude, moral obligation, incentive displacement).” In the Theory section (“Generalized Reciprocity”; “Reciprocity and the User Trajectory”) gratitude, moral obligation, and norm displacement are relabelled as proposed mechanisms the design cannot adjudicate. The Discussion subsection “The Response-Time Pattern and Its Interpretation” (Section 6.3) sets out the three competing readings of the response-time pattern — answer quality, re-engagement, and question characteristics — and advances all three as conjectures, and the Limitations section (Section 6.5) devotes a block to what the design cannot see, naming gratitude, moral obligation, incentive displacement, and re-engagement as <i>proposed</i> interpretations that our data do not identify. We have also renamed the session-dynamics account <i>re-engagement</i> throughout, so that a single mechanism no longer carries two labels.

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(Q1) <i>“The objectives are generally clear, but the rationale could be stated more modestly. The paper should clarify that its main contribution is not to establish a general theory of generalized reciprocity, but to test a short-run behavioral response to receiving an answer on Stack Overflow.”</i> (Q6) <i>“The tenure result is potentially the most interesting contribution and should receive more emphasis than the broader claims about moral reciprocity.”</i> (Q5) <i>“The conclusion should emphasize the narrower finding: receiving an answer is associated with a small short-run increase in helping, mainly among newer users.”</i>	We revised the abstract, introduction, and conclusion without changing any estimate. The Introduction now states in its opening paragraph that “our contribution is a cleaner short-run test on one platform: whether users help more after someone answers their own question,” and positions the estimates directly against prior empirical reports of online reciprocity (Baker 2014, Chen 2019 vs. Wasko 2005, Joyce 2006) in the “Empirical Challenges” subsection. The three-contribution paragraph makes the tenure gradient the headline (“the most robust and informative pattern in the data”), and the Conclusion (Section 7) now states exactly the narrower finding the reviewer prescribes — that the response “is strongest among newcomers and fades to nothing among the longest-tenured users” — and draws the reframing from it: generalized reciprocity works “less as a general engine of cooperation among established members than as a channel for recruiting contributors at the point of entry.” We report the supporting magnitudes (pooled arrival-increment HR 1.049, 95% CI [1.044, 1.054]; newcomer arrival HR 1.071) in the abstract and Results, and keep the Conclusion free of coefficients so that it carries the interpretation rather than repeating the tables. Consistent with this narrower, associational framing, we also retitled the manuscript from “Help Converts Newcomers, Not Veterans” to “At the Point of Entry: Receiving Help and Newcomer Contribution on Stack Overflow,” replacing the causal verb “converts” with associational language that matches the design while preserving the newcomer-centered emphasis.

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(Q2) “The authors should clearly distinguish questions, matched observations, interval rows, users, and help events. The reported sample sizes and event counts in the main tables are difficult to reconcile.” “The paper should explain more clearly how controls are duplicated or reused in matching, and how repeated observations from the same user are handled.” (Q4) “The paper should include clearer reporting of the estimation sample in each table: number of users, questions, matched observations, interval rows, and events where relevant.”	We rewrote the “Data” subsection as an explicit filtering-and-matching chain and added an appendix table that distinguishes exactly the five quantities the reviewer names. From the full non-deleted question set we removed 1,080 pre-launch and 8,029 post-cutoff cases and 2,671,121 self-answered questions, yielding 21,033,429 pre-matching questions from 4,764,048 users. Of these, 16,034,099 (76.2%) received at least one answer from another user and 4,999,330 (23.8%) received none — a split the previous draft never reported. Matching requires complete covariates within an exact stratum, leaving 12,290,847 answered and 3,345,139 unanswered questions eligible; of the eligible treated questions 12,174,880 (99.1%) were matched, 115,967 fell outside common support, and a further 3,743,252 answered questions lacked a valid stratum or complete covariates. We now state plainly that this attrition is <i>not</i> random: the treated propensity-score density piles up near one (Figure B.7), so the discarded questions are disproportionately those almost certain to be answered. On control reuse, the 12,174,880 matched treated questions draw on 2,481,556 <i>distinct</i> control questions, each reused about 4.9 times on average (up to 625) — the concrete answer to how controls are duplicated. New Appendix Table E.11 then maps users (3,732,706), questions (14,656,436), matched rows (24,349,760), interval rows (62,379,420), and events (3,435,973) per model, pooled and by tenure bucket, so the five quantities can no longer be confused. We also corrected a reporting ambiguity the reviewer’s concern exposed: the pooled <i>point estimates</i> use the full interval set, and only the matched-pair bootstrap replicates cap at approximately 8 million interval rows. A further reconciliation the earlier draft left implicit is now defined in “Implementation Details”: estimation rounds interval boundaries to a one-hour grid, under which same-hour helping events by a user merge, and this “time rounding” is what separates the raw descriptive event mean (0.18 per window, Table 4) from the estimation Events counts (0.1411 per row) and makes composite Events smaller than the sum of their channels. Finally, we sharpened the treatment/control definition itself (Variables subsection): treatment is a non-negative-scored answer from <i>another</i> user arriving within seven days, and controls are the complement — questions never answered, answered only by downvoted posts, or answered only after seven days. Every table derives from one pipeline run, so counts agree across the Data prose, Table 4, Table 5, and Table 7. All code and reproduction instructions are in the replication package (Data availability statement).

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(Q3) “Standard errors are a concern. The appendix states that robust or clustered standard errors are not used. Given repeated observations by users, repeated controls, matched pairs, and recurrent helping events, the current inference may be too optimistic.” “The authors should report robust standard errors clustered at least by user or matched pair, or use a bootstrap that preserves the matched structure.”	We implemented the second of the two remedies the reviewer offers — a bootstrap that preserves the matched structure — resampling whole matched pairs with replacement, refitting Model A, and taking percentile confidence intervals. (Native cluster-robust variances were not available to us in any case: <code>lifelines</code>’ <code>CoxTimeVaryingFitter</code> raises <code>NotImplementedError</code> for <code>robust=True</code> and exposes no cluster column for time-varying fits.) The bootstrapped statistic is the headline treatment effect itself — the difference-in-differences increment at answer arrival (β_4) — so the resampling uncertainty attaches directly to the quantity the paper reports. Table F.12 reports the pooled estimate (HR 1.049, bootstrap 95% CI [1.032, 1.075] at 100 replicates) alongside the model-based CI; the bootstrap interval is also shown in the pooled regression table (Table 5). Because the reviewer’s concern bites hardest where the paper’s claims are strongest, we extended the bootstrap to the <i>newcomer</i> (< 1 week) stratum — the estimate that carries the title, the abstract, and every design implication — at 500 replicates, with full-stratum refits and no per-replicate subsampling. The newcomer arrival increment returns a bootstrap 95% CI of [1.057, 1.084] (HR 1.071), which <i>confirms</i> the model-based interval [1.06, 1.09] rather than widening it. Preserving the matched structure therefore does not overturn the reported inference for the headline stratum, and Section 5 (H2 subsection) now states that this interval is bootstrap-confirmed. We should be explicit about one limit of this procedure, since the reviewer’s comment names repeated controls specifically: resampling whole pairs captures within- and between-pair dependence, but not the dependence induced by control <i>reuse</i>, because a control matched to several treated questions contributes the same pre-question baseline to each of them. We judge this residual to be second-order rather than negligible, for three reasons — the pair bootstrap already widens the pooled interval roughly fourfold relative to the model-based standard error, so it is capturing a substantial share of the dependence structure; each reuse is anchored to a different treated question’s response time, so the post-answer segments differ across reuses and the duplication is partial rather than exact; and the newcomer interval clears one by a wide margin. We therefore present these intervals as bootstrap-based rather than fully cluster-robust, and we flag the reused-control channel as the one source of dependence they do not absorb. Relatedly, we now report the waiting-period term (β_2) as an explicit parallel-trends diagnostic with its own standard error in every table, rather than folding it into any combined contrast: because β_2 carries no sign guarantee, a summed quantity such as $\exp(\beta_2 + \beta_4)$ bounds nothing, and we have

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(Q3) “Matching on activity history and tag-level answer rates helps, but the authors should add stronger robustness checks, such as controls for posting hour/day, question length, number of tags, user reputation, and text-based measures of question quality or complexity.” (Q7) “The paper should more clearly acknowledge that answer receipt is not random and may reflect unobserved question quality or user characteristics.”	We re-extracted fields dropped from the processed pipeline and added the requested observable pre-treatment covariates to the propensity model — posting hour-of-day and day-of-week, question body and title length (characters), and owner reputation — and now match pairs <i>exactly</i> on the number of tags in addition to the primary tag. Table 2 lists these in a new “Observable question and author characteristics” block, and Table 3 shows they are well balanced after matching (largest residual SMD 0.041, owner reputation; all below 0.05; the largest pre-matching imbalance is the tag-level accepted-answer rate, 0.695). In the pooled sample (Table G.14) and the newcomer stratum (Table G.16) the pooled arrival increment is essentially unchanged when these controls are additionally entered into the Cox model (1.04 with controls versus 1.05 baseline), and the newcomer increment is, if anything, strengthened (1.13 versus 1.07 baseline). We report one caution alongside that gain rather than resting on it: under the same controls the newcomer waiting-period diagnostic rises in step, from $\beta_2 = +0.050$ to $+0.122$ — essentially the magnitude of the arrival log-increment itself ($\log 1.13 = 0.121$). Both terms inflate together, so treated newcomers already diverge upward before any answer arrives in that specification. We therefore read it in the appendix as ruling out observable selection as the <i>sole</i> driver, not as independent corroboration of the arrival increment. The one requested control we did not add is a text-based measure of question quality or complexity beyond length: constructing a defensible complexity score for 21 million question bodies is outside what this revision could support, and the Discussion names text-based measures of question complexity or specificity as one of the things that would settle the response-time interpretation (Section 6.3). Limitations (Section 6.5) now sets out the identification position directly: the arrival increment “is identified under several assumptions, but one does the work: that nothing else which raises helping changes at that same moment,” and the section then separates the three kinds of selection that bear on it — observables, which the matching handles; stable unobservables, which the difference-in-differences removes; and time-varying unobservables, which neither device removes and which we cannot separate from reciprocity.

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(Q3) “The response-time analysis should be modeled more flexibly. The results appear closer to an inverted-U pattern than to a simple ‘faster is stronger’ effect.” (Q5) “H3 should be rewritten or toned down. The strongest effect occurs in the 30–60 minute bin, not among the fastest responses. This does not cleanly support the original hypothesis.” (Q4) “Figure 4 is useful, but the text should treat it as evidence of a non-linear response-time pattern rather than as simple support for faster responses.”	We replaced the single continuous log-response-time interaction with a flexible, non-parametric specification: ten discrete response-time bins, each fit as a separate Model A (Table 9, Figure 4). The bins are cut on the <i>actual</i> question-to-answer latency ($T_A - T_Q$), so “1–3 days” means answered 24–72 hours after posting; and because the observation window is anchored on the answer — it closes two days <i>after the answer arrives</i> ($T_A + 2D$), not at a fixed $T_Q + 2D$ — every treated question retains a full two-day post-answer phase and the arrival increment β_4 is identified in every bin, including the slow-answer bins. We revised Section 4 to state this answer-anchored window rule explicitly (the earlier draft’s “$T_Q + 2D$” phrasing described the design incorrectly) and added a note to Table 9 recording it. The pattern is non-linear — the arrival increment is near-null under 30 minutes, peaks sharply at 30–60 minutes (HR 1.19, 95% CI [1.17, 1.21]), and is modest thereafter, with a smaller secondary elevation at multi-day delays that we report but do not build on — an early “sweet spot,” not a monotone gradient. We rewrote the “Response Time Moderation (H3)” subsection around this specification, presenting the non-linear result first, and retitled the discussion subsection, presenting H3 as support for an immediacy account while noting that the fastest bins do not fit it. Acting on the reviewer’s specific instruction that H3 “should be rewritten or toned down,” we also rewrote the <i>hypothesis statement itself</i> (Section 2.3, “Response Time and the Immediacy of Reciprocity”), which previously asserted a strictly monotone relation (“Faster responses . . . strengthen the . . . effect, while slower responses weaken it”). It now reads: “<i>Timelier responses to a user’s question strengthen the generalized reciprocity effect.</i>” The paragraph that accompanies the hypothesis carries the qualification: laboratory accounts imply the advantage grows monotonically as help arrives sooner, field settings add session dynamics those accounts abstract away, and we therefore treat the prediction as directional and test whether it holds monotonically. The peak survives even strongly post-treatment answer-quality controls as the significant early-window maximum (1.03 [1.02, 1.04], $p < .001$; Table G.17; see R2 below).

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(Q4) “Table 5 should report hazard ratios and confidence intervals for both specifications, not only for the main model.”	We traced the missing interval to the table generator, not the model: the CI bounds for the speed specification’s treatment term were computed but dropped before the table was written. We added the read-out and regenerated the table. Table 5 now reports the coefficient, standard error, and HR with 95% CI for the arrival increment (β_4) in both columns, together with the waiting-period diagnostic (β_2) and its standard error. The only remaining em-dash is the Main+Speed bootstrap-CI cell, because the matched-pair bootstrap (Table F.12) was run for Model A alone.
(Q4; see also Reviewer 2, P7) “The authors should translate hazard ratios into absolute effect sizes, such as additional answers per 1,000 treated questions.” (Q7) “The small effect sizes should be discussed more explicitly.”	We added an absolute-effect derivation translating each tenure-bucket hazard ratio into an absolute rate difference (ARD), a number-needed-to-treat (NNT), and CI-derived NNT bounds, reported as a main-text table inside “Implications for Platform Design” (Table 10). Following the reviewer’s own phrasing, the table now carries an explicit <i>additional helping events per 1,000 answered questions</i> column. The newcomer arrival increment (HR 1.071) corresponds to 8.0 additional helping events per 1,000 answered newcomer questions (an absolute rate difference of 0.008 events per observation window), NNT 125.0 (95% CI 103.2–157.8); the pooled figure is 6.3 events per 1,000, NNT 158.4 (142.9–177.5). We also corrected the base rate these quantities are computed from: the earlier version divided by the event rate pooled across <i>both</i> arms, whereas the absolute effect of treatment must be referenced to the <i>control-arm</i> rate (0.1133 for newcomers, 0.1287 pooled), and the table note now states the $\text{ARD} = \text{control rate} \times (\text{HR} - 1)$ conversion together with its rare-event caveat. We also flag explicitly that the outcome is a recurrent-event <i>rate</i> rather than a binary risk, so NNT here is an approximate “answers needed per additional helping event” rather than a true number-needed-to-treat. Under the corrected base, newcomers show not only the largest relative response but among the best absolute yields (125 versus 126–128 for the one-to-twelve-month strata and 158 pooled), and the response is null among the most experienced users, so “Implications for Platform Design” was rewritten accordingly — foregrounding the modest magnitude and restricting the practical leverage to a targeted newcomer onboarding channel rather than a platform-wide multiplier.

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Reviewer comment	Response and changes to the manuscript
(Q7) <i>“The results are from Stack Overflow, a highly specific technical Q&A community, and may not generalize to other online communities.”</i>	Addressed jointly with Reviewer 2’s external-validity points below. The external-validity paragraph in Limitations (Section 6.5) now leads with Stack Overflow’s distinctiveness (large, relatively anonymous, gamified, technical) and refrains from claiming generalized reciprocity “likely operates” elsewhere, naming enterprise Slack channels, academic peer review, and intimate hobby forums as contrasting settings.
(Q8) <i>“The theory section should be shortened and more directly connected to the empirical design.”</i> <i>“Some implementation details currently in the appendix should be moved or summarized in the main text because they affect interpretation of the results.”</i>	We shortened the Theory section by roughly one-sixth (from about 1,550 to about 1,290 words), condensing the two-mechanism exposition of generalized reciprocity into a single paragraph (“Generalized Reciprocity” subsection) and compressing the empirical-challenges discussion. The theory now links forward to the design in two places. First, the empirical-challenges subsection closes: <i>“This confound is the central identification challenge that motivates the matched difference-in-differences design we develop in Section 4.”</i> Second, a new paragraph immediately after Hypothesis 2 (end of the “Reciprocity and the User Trajectory” subsection) carries the design stakes forward: <i>“If the reciprocity association is concentrated among newcomers, generalized reciprocity would function less as a general engine of cooperation among established members than as a contributor-recruitment channel operating at the point of entry, before community-specific incentives take hold — a design implication we develop in Section 6.4.”</i> We also promoted the tenure-stratified adoption figure from the appendix into the Results section (now Figure 6, immediately after the pooled Figure 5), together with its interpretive paragraph (“Response Time Moderation (H3)” subsection), so the disaggregated evidence bearing on the session-dynamics interpretation appears in the main text. The sample-construction and event-counting sub-point is addressed under Q2/Q8 above. One transparency note: the original appendix’s lifetime-contributions table was removed rather than promoted, because it derived from a superseded sample with an inconsistent data cutoff and constructs not defined in the current pipeline; no claim in the revised manuscript relies on it.

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(Q9) <i>Would you recommend language editing? — “Yes.”</i>	A full language and consistency edit was applied to the revised manuscript after the substantive revisions were complete. Representative corrections, with locations: “before and after an user receives an answer” now reads “... a user...” (opening of Section 4); a garbled sentence in the setting description now reads “ <i>Its user base spans a wide range of experience levels, from learners to professional software developers</i> ” (Section 3); a stray parenthesis and misplaced punctuation around citations were repaired (Section 3; “Analytical Strategy” subsection). Terminology and formatting were standardized throughout (“difference-in-differences,” defined as DiD at first use in the Introduction; “propensity score matching”; “response-time” as a modifier; “hazard ratio” defined at first use in both the abstract and the Introduction). The abstract was made self-contained by removing its cross-reference to an appendix table, and the Highlights were aligned with the revised abstract.

Response to Reviewer 2

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Reviewer comment	Response and changes to the manuscript
(Overall) <i>“The problem is that the topic is out-dated in the era of GAI. [...] This makes the topic not interesting.”</i> (P9) <i>“The paper treats its conclusions as timeless, but they are, at best, a pre-generative-AI baseline. The authors must add a dedicated limitations paragraph explicitly stating that their findings are a historical snapshot, not a prescriptive guide for contemporary or future platform design.”</i> (P2) <i>“The authors overgeneralise findings to all knowledge-sharing platforms, ignoring that Stack Overflow is uniquely technical, anonymous, and status-driven and critically, its ecosystem has been structurally disrupted by generative AI since 2023.”</i>	We revised “Empirical Setting: Stack Overflow” (Section 3) to remove the broad-generalization claim and to state that the platform’s active question-and-answer volume “has declined sharply since the release of generative AI tools in late 2022,” that the sample spans inception through 1 April 2025 (a period that includes but predates the steepest LLM-driven decline), and that the findings should be read as “a historical snapshot of one large, technical, relatively anonymous, status-driven platform.” These platform-decline claims, previously uncited, now carry supporting references (del Rio-Chanona et al. 2024; Burtch et al. 2024), and we corrected the platform statistics in the same passage (the “4.5 million users” figure was a miscitation; the site reports more than 29 million registered users, and the “mid-2025” statistic is now dated to June 2025, just after our 1 April 2025 cutoff). A dedicated Discussion subsection, “Generalized Reciprocity in the Generative-AI Era” (Section 6.6), states the window is a historical snapshot, not a prescriptive guide, and cross-references the pre-2020 robustness (Table G.13): both the pooled and newcomer associations are essentially unchanged on the pre-2020 subsample (pooled arrival HR 1.04 pre-2020 vs. 1.05 full sample; newcomer 1.07 in both), so the association predates rather than depends on the generative-AI era. We are careful not to oversell that check: because calendar year is an exact-matching stratum, the pre-2020 window is a large <i>sub-population</i> of the full sample rather than disjoint data (79% of matched rows, 91% of helping events), so we label it a compositional check rather than an independent replication. We further note that our own data register the decline the reviewer describes: the per-window helping rate is markedly lower after 2020 (about 0.057 versus 0.164 events per matched window before 2020). Finally, we did not stop at defensive framing: the Empirical Setting (Section 3) and the Conclusion now make the affirmative case for the topic’s timeliness — because the sample predates the steepest AI answer-substitution, it supplies a clean pre-substitution baseline that quantifies what a human answering ecosystem contributed to contributor recruitment while that ecosystem was intact, a measurement that may not be possible to make again. On the reviewer’s “contemporary or future” wording specifically, we have adopted it verbatim in the GAI-era paragraph and, to be clear about what we do and do not claim forward, we distinguish two senses in which the findings might bear on the future. We make no forward claim about the effect’s magnitude or its survival under AI substitution; on that our own post-2020 newcomer estimate is null and thin, and we say so. What the pre-substitution

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(P3) “The analytical sample only includes users who remained active long enough to post a question and be observed. Users who posted a question, received no answer, and immediately churned are excluded. Since this exclusion is likely correlated with the treatment (no answer) and the outcome (no future helping), the estimated treatment effect [...] is probably downward-biased. The authors should conduct a sensitivity analysis using Heckman-type selection correction or, at minimum, explicitly bound the possible bias.”	As a point of clarification about the sampling frame: the pipeline imposes no minimum-activity, minimum-tenure, or re-observation filter, so asked-but-never-answered users are <i>retained</i> as the control pool, and because each matched pair shares a fixed, pair-equalized observation window, an immediate churning simply contributes zero help events over a fully observed window. We did not implement a Heckman correction (no matching covariate provides a credible exclusion restriction, and the recurrent-event time-varying Cox is not natively Heckman-compatible). To characterize the exclusion the reviewer describes, we instead measured directly which questions the matching step discards (a descriptive check requiring no additional model fit): the exclusion is not one of user activity or tenure but of <i>common support</i>. Of the 4,999,330 unanswered questions in the pre-matching sample, 2,481,556 are used as matched controls; the remainder either fail the exact-stratum and complete-covariate requirement or are never selected as a nearest neighbour within the caliper, so roughly half the unanswered pool has no on-support match (the same accounting reported under Reviewer 1’s Q2 above). The retained control pool is accordingly drawn from more readily answered tags than the discarded tail (standardized mean difference of 0.66 on the tag-level accepted-answer rate). The estimand is thus the effect for answered questions with an on-support unanswered counterpart — a generalizability restriction inherent to matching rather than the activity-based selection the concern posits. Its direction, moreover, is the opposite of an upward bias: because the discarded questions are the <i>less</i> answerable ones, the more-answerable retained controls set a higher, not lower, comparison baseline, so the reported contrast is if anything conservative for the churn-prone population. Discarding is not concentrated in the newcomer stratum (it is dropped at close to its arm-wide average rate). The revised manuscript reports this accounting in Appendix Appendix C, which sets out the common-support restriction and its conservative direction, with a pointer to it from Limitations (Section 6.5). We treat the churn point itself as a clarification of the sampling frame rather than a limitation of the design, so we address it here rather than in the manuscript.

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(P4) “The outcome variable counts only ‘posting an answer’ as a helping event. On Stack Overflow, users frequently reciprocate through upvoting, editing, commenting, or accepting an answer. [These] behaviours require far less effort and are especially common among novices who lack expertise. By excluding these actions, the study systematically undercounts reciprocal behaviour and misclassifies many reciprocating users as non-responders. The authors should re-estimate the model using a composite helping indicator (including upvotes and comments) or justify why only answers capture the theoretically relevant construct, with reference to prior reciprocity literature.” (P1) “The operationalisation of ‘help’ is too narrow (answers only), excluding upvotes, edits, and comments — forms of low-cost reciprocity that newcomers often use.”	We did both of the things the reviewer proposes. First, the justification: a new “Why answers anchor the construct” paragraph (Variables subsection) grounds the answers-anchored outcome in the experimental reciprocity literature — generalized reciprocity is operationalized there as a <i>costly</i> prosocial contribution (Greiner & Levati 2005; Stanca 2009; Mujcic & Leibbrandt 2018), and it is the willingness to bear that cost, not low-cost approval, that distinguishes reciprocity from signaling (Molm et al. 2007). Second, the re-estimation: we parsed the additional event types from the raw dump and re-estimated the pooled model under every broader outcome definition the data support, reported as a full decomposition in Table 6: answers-only (headline) arrival HR 1.049 [1.044, 1.054]; comments-only 1.013 [1.007, 1.018]; edits-only 1.163 [1.155, 1.170]; answers+comments 1.024 [1.020, 1.029]; answers+comments+edits 1.045 [1.041, 1.050]. The decomposition shows the answers-only headline does not manufacture the effect: comments carry only a slight elevation, well below the answers increment, and adding them <i>dilutes</i> rather than amplifies the estimate — the opposite of the mechanical newcomer-undercount the reviewer is concerned about. Two of the channels the reviewer names cannot enter the model, for distinct reasons. Accepting an answer is a <i>self-directed</i> act: the asker marks an answer on their own question, and only treated users ever receive an answer to accept, so a matched control has no counterpart behavior and the difference-in-differences contrast is degenerate rather than informative — we note this where the decomposition is introduced. Up-votes are simply unobservable: the public data dump does not record who cast a vote — its up-vote records carry no voter <code>UserId</code> — so an “up-votes given” signal cannot be constructed for any user from Stack Overflow’s released data at all. This is a property of the data source, not of our processing: re-parsing the raw dump would not recover the information, because it was never released. Limitations (Section 6.5) opens on what we measure: the outcome is “the posting of an answer to another user’s question, which we treat as costly public contribution,” and reciprocity “can also run through channels we do not observe” — up-votes on the platform, where the dump withholds the voter’s identity, and helping off the platform entirely. The section also notes the direction of that omission: if newcomers reciprocate through those unobserved channels more than veterans do, the reported tenure gradient is conservative.

Reviewer comment	Response and changes to the manuscript
(P5) <i>“The paper interprets the peak reciprocity at 30–60 minutes as a ‘re-engagement window’ effect. However, a simpler alternative is answer quality: extremely fast answers (<15 min) are often terse, link-only, or low-scoring, while 30–60-minute answers are typically more detailed, with code snippets and explanations, generating greater gratitude. The authors do not control for answer length, vote score, or whether the answer was accepted. Without testing this confound (e.g., by conditioning on answer quality metrics), the structural ‘session-dynamics’ interpretation remains speculative and should be toned down.”</i>	We re-extracted three observable quality signals (first-answer length in characters, vote score, and acceptance) and re-estimated the pooled speed specification, adding them one at a time as a graded decomposition (Table G.15). Controlling for <i>answer length</i> alone — the reviewer’s terse-versus-detailed variable, and the signal least entangled with the asker’s own behavior — barely moves the estimate (arrival 1.038 to 1.032); vote score alone brings it to 1.010; and the full specification, which additionally conditions on acceptance, pushes it below one (0.982), with the same pattern in the newcomer stratum (quality-controlled arrival 0.95; Table G.16). Because vote score and, especially, acceptance are realized only for answered questions and are endogenous to the outcome (accepting requires the asker to return to the site), we frame these level estimates as post-treatment bounds rather than estimates of the effect net of quality, report them in the appendix (Tables G.15 and G.18), and keep the bin-level test — the direct test of the reviewer’s timing-versus-quality concern — in the main text (“Answer quality as an alternative explanation”). The session-dynamics reading is recast as one conjecture among several (quality, session dynamics, selection). We also verified that the below-one result is not an artifact of how the vote score is coded: re-estimating the score-only and full specifications under a signed-log (asinh) coding and ordinal-bin codings of the score (at full data, pooled and in the newcomer stratum) leaves the full-specification increment below one under all three codings, with the better-specified signed-log and binned codings if anything pulling it marginally <i>further</i> below one; the only estimate sensitive to the coding is the intermediate score-only step, which is above one under the linear coding but statistically indistinguishable from one under the alternatives (new Appendix Table G.18). We use $\text{asinh}(\text{score})$ rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative (downvoted) scores, for which $\log(1 + \text{score})$ is undefined.

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Reviewer comment	Response and changes to the manuscript
<i>(P5, continued) “The striking inverted-U response-time effect (peak at 30–60 minutes) conflates answer speed with answer quality; faster answers may be terse or incorrect, while 30-minute answers tend to be more thorough, a confound not tested.”</i>	Beyond the pooled test above, we re-estimated each of the ten discrete response-time bins with the same three observable quality controls, fitting Model A plus those signals per bin so the baseline and quality columns are directly comparable (Table G.17). The 30–60-minute peak attenuates from 1.19 [1.17, 1.21] to 1.03 [1.02, 1.04] but remains the clearly significant ($p < .001$) early-window maximum, while the same controls push the fastest bins significantly <i>below</i> one (0.95) — a signature of over-adjustment, since score and acceptance are strongly post-treatment (acceptance requires the asker’s return, part of the re-engagement pathway that produces the outcome). That the peak survives an adjustment biased against it indicates that answer timing retains explanatory content beyond observable quality. “Response Time Moderation (H3)” reports this as genuine but bounded support for the immediacy account; a persistent secondary elevation at one to three days (1.09) is left uninterpreted because the controls are post-treatment.

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Reviewer comment	Response and changes to the manuscript
(P6) <i>“The platform has undergone massive cultural, technical, and policy shifts from 2008 to 2025 — including reputation inflation, spam filters, tag evolution, and most critically, the rise of ChatGPT post-2022. [...] the tenure-stratified analysis pools newcomers from 2008 with those from 2024 as if they share the same ‘newcomer’ experience. This ignores cohort heterogeneity. The authors should include year-of-entry fixed effects or conduct robustness checks on pre-2020 subsamples to isolate the pure human-only reciprocity mechanism.”</i>	A year-of-entry fixed-effects specification would be collinear with our matched design — calendar year already enters as an exact-matching stratum, so registration cohort is absorbed once question-year is matched — so we instead pursued the pre-2020 subsample the reviewer also proposed, both pooled and in the newcomer stratum. Re-estimating Model A before 2020 (Table G.13) leaves the arrival increment essentially unchanged (pre-2020 pooled arrival HR 1.04 versus 1.05 full sample; pre-2020 newcomer 1.07 versus 1.07 full sample). Because that subsample retains 79% of matched rows and 91% of helping events, however, we now describe it as a compositional check rather than an independent replication, and we went beyond the reviewer’s request by also estimating the <i>post-2020</i> window separately — a partition that shares no rows with the pre-2020 check. We report both outcomes, including the less convenient one. The pooled arrival increment replicates across eras (post-2020 HR 1.04, 95% CI [1.03, 1.06]). The newcomer-specific estimate, by contrast, is not distinguishable from one after 2020 (HR 1.00, [0.97, 1.03]) and rests on far fewer events (31,814 versus 336,966 pre-2020), which is consistent with the post-2022 contraction in newcomer activity the reviewer describes and which Burtch et al. (2024) document. We therefore state in the “Experience Moderation of Reciprocity (H2)” subsection that the newcomer association is well identified in the pre- and early-generative-AI data that dominate our sample rather than in the sparser post-2020 period, and we frame the contribution accordingly: a pre-substitution baseline for what human answering did for contributor recruitment, not a claim about the platform’s current state. The Limitations additionally offer two labeled conjectures for the attenuation — cohort selection among post-2022 joiners, and a residual question mix that skews beyond a newcomer’s ability — and mark both as untestable with our archival data.

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Reviewer comment	Response and changes to the manuscript
(P7) <i>“The paper does not report the Number Needed to Treat (NNT) — i.e., how many newcomers must receive an answer to generate one additional helping event. Without this, platform designers cannot assess whether investing in faster response systems yields meaningful returns. The authors should calculate and report NNT, and revise their ‘Implications for Design’ section to reflect modest effect sizes.”</i>	Addressed jointly with Reviewer 1’s absolute-effects request above. Table 10 — placed in the main text inside the revised “Implications for Platform Design” so the absolute magnitudes sit next to the design claims they discipline — reports the absolute rate difference, NNT, NNT confidence bounds, and additional helping events per 1,000 answered questions by tenure bucket. Answering roughly 125 newcomer questions yields one additional helping event (95% CI 103.2–157.8), equivalently 8.0 events per 1,000 answered newcomer questions; the pooled figure is NNT 158.4 (142.9–177.5), and the response is null among the most experienced users. These quantities are referenced to the <i>control-arm</i> event rate (0.1133 newcomers, 0.1287 pooled) rather than the both-arms pooled rate used in the previous version, and the note states the conversion and its approximation. We also flag that the outcome is a recurrent-event rate rather than a binary risk, so this NNT is an approximate “answers needed per additional helping event.” “Implications for Platform Design” foregrounds the modest absolute magnitude and, consistent with the reviewer’s point, declines to recommend interventions built specifically on response speed.

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Reviewer comment	Response and changes to the manuscript
(P8) “Posting a question increases a user’s profile visibility: other members click on their profile, view their history, and may trigger a sense of social obligation or reputation concern. This ‘exposure effect’ is not equivalent to gratitude for receiving an answer, yet it operates precisely during the post-answer period and is not separately identified. The waiting-period covariate absorbs baseline question-asking effects, but it does not capture the increased visibility that persists after an answer arrives. A placebo test using questions that received no answers but high views could help rule out this confound; currently, it is not attempted.”	We agree that post-answer exposure is a real threat to mechanism <i>attribution</i>, and we now treat it explicitly in Limitations (Section 6.5); on reflection, however, the <i>direction</i> of this bias is not established, and we have revised the manuscript so that it no longer presumes exposure inflates the arrival increment. The exposure the reviewer describes must be carried by contemporaneous, logged-in users who browse question feeds and click through to profiles — the same population that supplies potential answerers — and for those users receiving an answer tends to <i>lower</i> a question’s visibility: once a question carries an upvoted or accepted answer it leaves the “unanswered” queue that answerers browse and is marked resolved, so relative to still-unanswered matched controls it draws fewer such viewers. Under our matched difference-in-differences this channel would therefore <i>attenuate</i> the arrival increment, making our estimate conservative rather than inflated; the one component that clearly runs the other way is transient, a new answer briefly bumping the question toward the top of activity-sorted feeds in the hours after it arrives. Because these components run in opposite directions and neither is observable in the archival dump, Limitations now states that the channel operates “though not in an obvious direction”: being answered “lifts a question in activity-sorted feeds for a while, but it also takes the question out of the unanswered queue that potential answerers browse, and the second effect persists while the first is transient.” The manuscript draws the implication the reviewer’s framing does not anticipate — “if anything, then, this channel works against the increment we report rather than inflating it” — so the concern, followed through, makes our estimate conservative rather than inflated. We have also given the channel a name in the manuscript — <i>social expectation</i> — and set out the two conditions under which it would explain our results away: visibility must rise when the answer arrives rather than when the question is posted, and the rise must bind newcomers more tightly than veterans. The first is plausible; the second is where the account strains, since veterans have more reputation at stake and more visible profiles yet do not respond at all. This also sharpens why the proposed placebo is not identified: the dump records <i>ViewCount</i> only as a single cumulative snapshot at the dump date, and that count is dominated by later, largely logged-out search-engine visitors who never reach a profile — the wrong population for the exposure construct — so it conflates a question’s age, topic popularity, and years of downstream traffic with the contemporaneous answerer attention such a placebo would require; testing the channel directly would need time-stamped view data or repeated snapshots that do not exist for our study period. Finally, we note that the

Reviewer 2 (continued)

Reviewer comment	Response and changes to the manuscript
<i>(Q9) Would you recommend language editing? — “Yes.”</i>	Addressed jointly with Reviewer 1’s Q9 above: a full language and consistency edit was applied to the revised manuscript after the substantive revisions were complete.

We hope these revisions address the reviewers’ concerns. We are grateful for feedback that materially improved the identification, the transparency of the sample and inference, and the calibration of our claims, and we are happy to provide any further analyses the reviewers consider useful.